Mode-Aware Anti-Jamming for Dynamic Spectrum Sharing UAV Networks:Joint Spectrum Allocation, Trajectory, and Power Optimization

***Abstract*—Unmanned aerial vehicle (UAV) communication systems operating under dynamic spectrum sharing (DSS) are highly vulnerable to intentional interference in adversarial environments. In practice, jammers often adopt pattern-switching strategies, alternating among multiple jamming modes such as constant, sweep, and random interference, which introduces abrupt and non-stationary interference dynamics. Such behavior significantly degrades the quality-of-service (QoS) of UAV-assisted communications and makes rapid recovery after interference transitions particularly challenging. To address this problem, this paper proposes a mode-aware anti-jamming framework for DSS-enabled UAV networks. A CNN–GRU based jamming mode recognition module is first designed to infer the jammer's operating pattern from short-term interference observations. The inferred mode probability is then incorporated into a semantic-augmented Markov decision process (MDP). Based on this formulation, a hybrid reinforcement learning framework combining proximal policy optimization (PPO) and deep deterministic policy gradient (DDPG) is developed to jointly optimize spectrum allocation, UAV trajectory, and transmit power. Simulation results demonstrate that the proposed approach achieves faster recovery after jamming mode transitions and improves both communication reliability and spectrum utilization compared with existing schemes.**

***Index Terms*—**

I. Introduction

Unmanned aerial vehicles (UAVs) have emerged as a promising solution for providing flexible and on-demand wireless connectivity in low-altitude networks. By exploiting their high mobility and line-of-sight communication links, UAV-assisted systems can effectively support emergency communications, temporary hotspots, and aerial relaying services. However, due to the scarcity of dedicated spectrum resources, UAV networks are often required to operate under dynamic spectrum sharing (DSS) frameworks, where secondary UAV links opportunistically access licensed spectrum while coexisting with incumbent primary users (PUs).

Although DSS significantly improves spectrum utilization efficiency, it also exposes UAV communications to highly dynamic interference environments. In particular, adversarial jammers can intentionally disrupt wireless transmissions by injecting strong interference signals. In practice, such jammers rarely adopt a fixed interference strategy; instead, they frequently switch among multiple jamming patterns, such as constant, sweep, and random jamming, in order to increase uncertainty and degrade adaptive communication performance. This pattern-switching jamming behavior introduces abrupt and non-stationary interference dynamics, which may lead to severe quality-of-service (QoS) degradation and communication outages for UAV-assisted secondary links. For mission-critical UAV applications, rapidly recovering communication performance after interference transitions is often more important than merely optimizing long-term average throughput.

To improve spectrum efficiency and communication reliability, extensive research has investigated joint spectrum allocation and UAV trajectory optimization in DSS-enabled UAV networks. By leveraging the UAV's mobility and flexible spectrum access capability, learning-based approaches can dynamically adapt resource allocation and flight control in complex environments. In particular, reinforcement learning (RL) has been widely applied to address the coupled discrete–continuous optimization problem arising from spectrum assignment and UAV movement. For example, recent works combine actor–critic learning with deep deterministic policy gradient (DDPG) to jointly optimize spectrum allocation and UAV trajectory in dynamic spectrum sharing networks.

However, most existing methods treat interference as an exogenous disturbance and directly incorporate raw interference measurements into the state space. The learning agent is therefore expected to implicitly infer jammer behavior from long-term reward feedback. Under pattern-switching jamming, this implicit learning mechanism leads to an effectively non-stationary environment, where the learned policy tends to converge to conservative average-case behaviors and exhibits slow adaptation when the interference pattern changes. Moreover, existing designs rarely consider explicit jamming cognition or adaptive transmit power control, which further limits the system's capability to rapidly recover link quality after sudden interference transitions.

To address these challenges, this paper investigates mode-aware anti-jamming communication for DSS-enabled UAV networks under pattern-switching jamming. We introduce an explicit jamming cognition mechanism that enables the UAV to recognize the current interference pattern and incorporate this semantic information into the decision-making process. Based on the recognized jamming mode, a hybrid reinforcement learning framework is proposed to jointly optimize spectrum allocation, UAV trajectory, and transmit power, aiming to achieve fast communication recovery and reliable spectrum sharing in adversarial environments.

The main contributions of this work are summarized as follows:

- We develop a mode-aware anti-jamming framework for

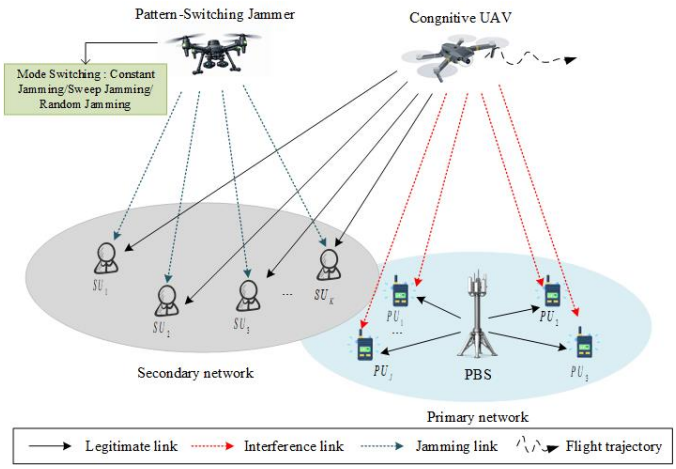

Fig. 1. Dynamic spectrum sharing UAV network under pattern-switching jamming, where the C-UAV performs jamming mode awareness and jointly optimizes spectrum allocation, trajectory, and transmit power to ensure reliable communication.

UAV-enabled DSS networks with pattern-switching jammers, where the UAV dynamically senses and interprets interference behaviors.

• A CNN–GRU based jamming mode recognition module is designed to infer the jammer's operating pattern from short-term interference observations and provide semantic information for decision making.

• We formulate a semantic-augmented Markov decision process (MDP) and propose a hybrid PPO–DDPG learning framework that jointly optimizes spectrum allocation, UAV trajectory, and transmit power.

• Simulation results demonstrate that the proposed method achieves faster recovery after interference transitions, higher average throughput, and lower outage probability compared with existing approaches.

II. System Model

A. Scenario Description

We consider a UAV-enabled dynamic spectrum sharing (DSS) network under intentional pattern-switching jamming, as illustrated in Fig. 1. The network consists of a primary network and a cognitive UAV-assisted secondary network, which coexist over a set of licensed spectrum bands.

The primary network includes a primary base station and J primary users (PUs). The PUs are indexed by $j \in \mathcal{J} \triangleq \{1, 2, \ldots, J\}$. In order to serve more secondary users (SUs) and provide better services, a wideband spectrum is divided into M sub-carrier spectrum bands. The spectrum bands are indexed by $m \in \mathcal{M} \triangleq \{1, 2, \ldots, M\}$. To eliminate mutual interference among PUs, each PU is assigned an exclusive licensed spectrum band. The secondary network comprises a cognitive UAV serving K ground SUs. The SUs are indexed by $k \in \mathcal{K} \triangleq \{1, 2, \ldots, K\}$. The C-UAV operates as an aerial secondary base station, providing downlink communications to SUs by opportunistically accessing the licensed spectrum while ensuring that the interference inflicted on the primary network remains within acceptable limits.

A three-dimensional Cartesian coordinate system is adopted. The horizontal locations of the PBS, the j-th PU, the k-th SU, the C-UAV and the J-UAV at time slot n are denoted by $\mathbf{w}_b = (x_b, y_b)$, $\mathbf{w}_{p,j} = (x_{p,j}, y_{p,j})$, $\mathbf{w}_{s,k} = (x_{s,k}, y_{s,k})$, $\mathbf{q}_c = (x_c, y_c)$ and $\mathbf{q}_{jam} = (x_{jam}, y_{jam})$ respectively. These UAVs fly at a fixed altitude H_c. The total communication duration T is divided into N equal-length time slots, each with duration $\Delta t = T/N$. Due to the short duration of each slot, the C-UAV is assumed to be quasi-static within one slot. Let $n \in \mathcal{N} \triangleq \{1, 2, \ldots, N\}$ denote the set of time slots. The dynamic positions of the SUs can be formulated as

$$\begin{aligned} x_{s,k}[n+1] &= x_{s,k}[n] + v_s \cos\big(\phi_{s,k}[n]\big), \\ y_{s,k}[n+1] &= y_{s,k}[n] + v_s \sin\big(\phi_{s,k}[n]\big), \end{aligned} \tag{1}$$

where $\phi_{s,k}[n] \in [-\pi, \pi]$ denotes the direction of the k-th SU at time slot n. The SUs move at a fixed speed v_s. Similarly, the dynamic position of the C-UAV can be formulated as

$$\begin{aligned} x_c[n+1] &= x_c[n] + v_c \cos\big(\phi_c[n]\big), \\ y_c[n+1] &= y_c[n] + v_c \sin\big(\phi_c[n]\big), \end{aligned} \tag{2}$$

where v_c is the constant flight speed, and $\phi_c[n] \in [-\pi, \pi]$ denotes the direction of the C-UAV. The distance between the C-UAV and the k-th SU, and that between the PBS and the j-th PU at time slot n are respectively given as,

$$\begin{aligned} d_{c,k}[n] &= \sqrt{\big\|\mathbf{q}_c[n] - \mathbf{w}_{s,k}[n]\big\|^2 + H_c^2}, \\ d_{p,j}[n] &= \sqrt{\big\|\mathbf{w}_b[n] - \mathbf{w}_{p,j}[n]\big\|^2}, \end{aligned} \tag{3}$$

The wireless channel between the UAV and the ground users is dominated by the line-of-sight (LoS) link. Let β_{ref} represent the channel power gain at the reference distance of 1 m. Thus, the channel power gain from the C-UAV to the k-th SU can be expressed as

$$h_{C,k,n}^{\text{LoS}} = \beta_{\text{ref}} d_{c,k}[n]^{-2}, \tag{4}$$

where $d_{c,k}[n]$ denotes the distance between the C-UAV and the k-th SU at time slot n. The channel model between the PBS and the ground users is different from that between the UAV and the ground users.It is required to consider both the distance-dependent path loss with exponent $\varphi \geq 2$ and small-scale Rayleigh fading. Thus, the channel power gains from the PBS to the j-th PU can be given as

$$h_{P,j,n}^{\text{NLoS}} = \beta_{\text{ref}} d_{p,j}[n]^{-\varphi} \zeta_j, \tag{5}$$

where $d_{p,j}[n]$ is the distance between the PBS and the j-th PU at time slot n. The random variable ζ_j follows an exponential distribution with unit mean, which accounts for the Rayleigh fading.

Unlike conventional fixed or random jamming models, the J-UAV is assumed to adopt a pattern-switching strategy. Let $z[n] \in \mathcal{Z} \triangleq \{\text{Constant}, \text{Sweep}, \text{Random}\}$ denote the jamming mode at time slot n, corresponding to constant jamming, sweep jamming, and random jamming, respectively.

The jamming mode evolves over time according to a finite-state stochastic process. In particular, the mode transition is modeled as a first-order Markov chain, i.e.,

$$\Pr(z[n+1] = z' \mid z[n] = z) = \Pi_{z,z'}, \quad z, z' \in \mathcal{Z}. \tag{6}$$

where $\mathbf{\Pi}$ denotes the mode transition probability matrix. The transition probabilities are not known to the C-UAV, and the instantaneous jamming mode cannot be directly observed. The pattern-switching behavior of the jammer directly translates into time-varying and non-stationary interference across the spectrum.

Considering the sub-carrier allocation, the binary variables $\rho_{k,n}[m]$ and $\rho_{J,n}[m]$ are introduced to characterize the spectrum usage of the SUs and the J-UAV, respectively.Specifically, $\rho_{k,n}[m] = 1$ indicates that the m-th sub-carrier is assigned to the k-th SU at time slot n, and $\rho_{k,n}[m] = 0$ otherwise. Each PU operates on a preassigned orthogonal sub-carrier with a fixed transmit power. The J-UAV dynamically occupies sub-carriers according to its current jamming mode, which is captured by the jamming indicator $\rho_{J,n}[m]$. When $\rho_{J,n}[m] = 1$, the m-th sub-carrier is subject to intentional interference from the jammer at time slot n. Based on the sub-carrier availability and the time-varying jamming condition, the C-UAV aims to dynamically adapt the spectrum allocation for the SUs, such that the communication quality can be rapidly restored after jamming pattern changes, while opportunistically accessing the licensed spectrum and guaranteeing the quality-of-service requirements of the primary network.

The SINR of the k-th SU and the j-th PU in the m-th sub-band can be respectively expressed as

$$\gamma^{s}_{m,k,n} = \frac{\rho_{k,n}[m] P_C[n] h^{\mathrm{LoS}}_{C,k,n}}{\sigma^2 + P_B h^{\mathrm{NLoS}}_{P,k,n} + \rho_{J,n}[m] P_J h^{\mathrm{LoS}}_{J,k,n}}, \tag{7a}$$

$$\gamma^{p}_{m,j,n} = \frac{P_B h^{\mathrm{NLoS}}_{P,j,n}}{\sigma^2 + \sum_{k=1}^{K} \rho_{k,n}[m] P_C[n] h^{\mathrm{LoS}}_{C,j,n} + \rho_{J,n}[m] P_J h^{\mathrm{LoS}}_{J,j,n}}, \tag{7b}$$

where P_B and P_J denote the transmit power of the PBS and the J-UAV, respectively. $P_C[n] \in [P_C^{\min}, P_C^{\max}]$ denotes the transmit power of the C-UAV at time slot n. Moreover, σ^2 is the noise power. Then, the achievable data transmission rate of the k-th SU and the j-th PU can be respectively expressed as

$$R^{s}_{m,k,n} = B \log_2\left(1 + \gamma^{s}_{m,k,n}\right), \tag{8a}$$

$$R^{p}_{m,j,n} = B \log_2\left(1 + \gamma^{p}_{m,j,n}\right), \tag{8b}$$

where B denotes the bandwidth for each sub-band.

To capture communication reliability under pattern-switching jamming, we introduce a QoS threshold R_{th}. An outage event is declared when the achievable rate of a SU falls below R_{th}. Accordingly, the outage indicator is defined as

$$\mathbb{I}^{\mathrm{out}}_{k,n} \triangleq \mathbf{1}\left\{R^{s}_{k,n} < R_{\mathrm{th}}\right\}, \quad \forall k \in \mathcal{K},\ \forall n \in \mathcal{N}, \tag{9}$$

where $\mathbf{1}\{\cdot\}$ equals 1 if the condition holds and 0 otherwise. This definition penalizes each outage time slot; thus, longer outage durations yield larger cumulative penalties, which encourages fast recovery after jamming pattern transitions.

B. Problem Formulation

In this paper, we address a joint spectrum allocation, UAV trajectory control, and transmit power adaptation problem in a dynamic spectrum sharing UAV network under pattern-switching jamming, aiming to achieve fast recovery and reliable communication for secondary users while satisfying the primary user protection constraints. Accordingly, the optimization problem is formulated as follows

$$\textbf{P1:} \quad \max_{\boldsymbol{\rho}, \boldsymbol{\phi}, \mathbf{P}} \quad \frac{1}{N} \sum_{n=1}^{N} \sum_{k=1}^{K} \sum_{m=1}^{M} R^{\mathrm{s}}_{m,k,n} - \frac{\lambda_{\mathrm{out}}}{N} \sum_{n=1}^{N} \sum_{k=1}^{K} \mathbb{I}^{\mathrm{out}}_{k,n} \tag{10a}$$

$$\text{s.t. } \textbf{C1:} \quad \sum_{m=1}^{M} R^{p}_{m,j,n} \geq R^{\min}_{j}, \quad \forall j \in \mathcal{J},\ \forall n \in \mathcal{N} \tag{10b}$$

$$\textbf{C2:} \quad \rho_{k,n}[m] \in \{0,1\}, \quad \forall k \in \mathcal{K},\ \forall m \in \mathcal{M},\ \forall n \in \mathcal{N} \tag{10c}$$

$$\textbf{C3:} \quad \sum_{k=1}^{K} \rho_{k,n}[m] \leq 1, \quad \forall m \in \mathcal{M},\ \forall n \in \mathcal{N} \tag{10d}$$

$$\textbf{C4:} \quad |\phi_c[n]| \leq \pi, \quad \forall n \in \mathcal{N} \tag{10e}$$

$$\textbf{C5:} \quad P_C^{\min} \leq P_C[n] \leq P_C^{\max}, \quad \forall n \in \mathcal{N} \tag{10f}$$

To jointly account for throughput performance and communication reliability, a reliability-aware objective function is adopted. Specifically, the achievable rate of the secondary users is maximized, while a penalty term is introduced to capture QoS outages. This outage penalty discourages prolonged service interruptions caused by dynamic jamming and implicitly promotes fast recovery of communication quality after jamming pattern transitions. In problem (P1), $R^{\min}_{j}$ denotes the minimum transmission rate requirement of the j-th PU, and $\boldsymbol{\rho} = \{\rho_{k,n}[m]\}_{k\in\mathcal{K}, m\in\mathcal{M}, n\in\mathcal{N}}$ denotes the spectrum allocation policy for the SUs. The variable $\boldsymbol{\phi} = \{\phi[n]\}_{n\in\mathcal{N}}$ denotes the UAV heading angle control sequence, while $\mathbf{P} = \{P_C[n]\}_{n\in\mathcal{N}}$ denotes the transmit power adaptation policy of the cognitive UAV. Constraint (10b) guarantees the QoS requirements of the PUs in the dynamic spectrum sharing environment. Constraints (10c) and (10d) ensure orthogonal sub-carrier allocation among SUs to avoid intra-secondary interference. Constraint (10e) enforces feasible UAV mobility, while constraint (10f) restricts the transmit power of the C-UAV within practical limits.

III. Our Proposed Mode-Aware Anti-Jamming Framework

A. Jamming Mode Recognition

The jammer switches among finite modes $z[n] \in \mathcal{Z}$ following the Markov model in (6). The C-UAV performs spectrum sensing and collects the per-slot interference-energy vector

$$\mathbf{I}[n] = [I_1[n], \ldots, I_M[n]] \in \mathbb{R}^M, \tag{11}$$

and forms a length-L short window

$$\mathcal{I}[n] = \{\mathbf{I}[n-L+1], \ldots, \mathbf{I}[n]\} \in \mathbb{R}^{L\times M}. \tag{12}$$

A lightweight recognizer $f_\theta(\cdot)$ maps the input $\mathcal{I}[n]$ to a mode probability vector

$$\boldsymbol{\pi}[n] = f_\theta(\mathcal{I}[n]) \in \mathbb{R}^{|\mathcal{Z}|}, \quad \sum_{c=1}^{|\mathcal{Z}|} \pi_c[n] = 1, \tag{13}$$

which provides semantic information about the current jamming behavior. We adopt a CNN–GRU classifier, where a 1D CNN encodes the spectral characteristics of each $\mathbf{I}[\cdot]$, and a GRU captures short-term temporal dynamics over L time slots. The model is trained offline using the cross-entropy loss

$$\mathcal{L}_{\text{cls}}(\theta) = -\sum_{c=1}^{|\mathcal{Z}|} z_c[n] \log\big(\pi_c[n]\big), \tag{14}$$

where $z_c[n]$ denotes the one-hot encoded ground-truth jamming mode at time slot n. During online operation, the model is used solely for inference to output $\boldsymbol{\pi}[n]$.

B. Semantic-Augmented MDP Formulation

Following the MDP formulation in the reference work [1], we incorporate the mode probability vector $\boldsymbol{\pi}[n]$ as a semantic descriptor of the interference environment to facilitate fast adaptation under mode-switching jamming.

State: The system state at slot n is defined as

$$\mathbf{s}[n] = \big(\mathbf{a}[n-1],\, \mathbf{q}_c[n-1],\, \mathbf{g}[n],\, \boldsymbol{\pi}[n]\big), \tag{15}$$

where $\mathbf{a}[n-1]$ denotes the previous action at slot $n-1$, $\mathbf{q}_c[n-1]$ denotes the previous UAV position at slot $n-1$, $\mathbf{g}[n]$ denotes channel and geometry-related features for rate evaluation, and $\boldsymbol{\pi}[n]$ encodes the high-level semantic belief of the current jamming behavior.

Action: The action consists of a discrete spectrum decision and a continuous control vector,

$$\mathbf{a}[n] = \big(\mathbf{a}_d[n],\, \mathbf{a}_c[n]\big), \tag{16}$$

The discrete action $\mathbf{a}_d[n]$ represents a joint subcarrier allocation for all K SUs at slot n, i.e.,

$$\mathbf{a}_d[n] = \big(a_1[n],\, a_2[n],\, \ldots,\, a_K[n]\big), \tag{17}$$

where $a_k[n] \in \{1, 2, \ldots, M\}$ denotes the selected subcarrier index for the k-th SU. Equivalently, $\mathbf{a}_d[n]$ can be mapped to the binary allocation variables $\{\rho_{k,n}[m]\}$ satisfying $\sum_{m=1}^{M} \rho_{k,n}[m] = 1$ for each k. The continuous action $\mathbf{a}_c[n]$ is defined as

$$\mathbf{a}_c[n] = \big(\phi_c[n],\, P_C[n]\big), \tag{18}$$

where $\phi_c[n] \in [-\pi, \pi]$ and $P_C[n] \in [P_C^{\min},\, P_C^{\max}]$.

Reward: Consistent with (10), we adopt a reliability-aware reward that jointly captures the secondary throughput, outage reliability, and PU protection, defined as

$$r[n] = \alpha \sum_{k=1}^{K} R_k^s[n] + \beta \sum_{j=1}^{J} \delta_j[n] - \gamma \sum_{k=1}^{K} \mathbb{I}(R_k^s[n] < R_{\text{th}}), \tag{19}$$

where $\alpha, \beta, \gamma > 0$ are constant coefficients, $\delta_j[n]$ is a penalty term when the interference to the PUs exceeds the tolerated threshold and formulated as

$$\delta_j[n] = \begin{cases} 0, & R_j^p[n] \geq R_j^{\min}, \\ R_j^p[n] - R_j^{\min}, & 0 \leq R_j^p[n] < R_j^{\min}, \end{cases} \tag{20}$$

where $R_j^p[n]$ is the achievable rate of the j-th PU and $R_j^{\min}$ is its minimum rate requirement.

C. Hybrid PPO-DDPG Framework

Problem (10) involves discrete spectrum allocation, continuous joint control of UAV mobility and transmit power. Thus, we employ a hybrid solver, where PPO is adopted for the discrete spectrum decision $\mathbf{a}_d[n]$ and DDPG is used for the continuous control vector $\mathbf{a}_c[n]$, both conditioned on the semantic-augmented state $\mathbf{s}[n]$.

PPO for Spectrum Allocation: Let $\pi_\theta(\mathbf{a}_d|\mathbf{s})$ denote the stochastic policy for spectrum allocation parameterized by θ, where $\mathbf{a}_d[n]$ is the discrete joint subcarrier allocation decision at time slot n. Different from the multi-agent A2C design in [1], we adopt a centralized PPO-based spectrum allocator. To avoid the exponential growth of the joint discrete action space, the policy is implemented in a factorized multi-discrete form as

$$\pi_\theta(\mathbf{a}_d|\mathbf{s}) = \prod_{k=1}^{K} \pi_\theta^{(k)}\big(a_k|\mathbf{s}\big), \tag{21}$$

where each $\pi_\theta^{(k)}(\cdot|\mathbf{s})$ is a categorical distribution over the M candidate subcarriers. To satisfy the orthogonal subcarrier constraint, action masking is adopted during action selection to prevent infeasible allocations and ensure that each subcarrier is assigned to at most one SU per slot. The objective of PPO is to maximize the expected discounted return while preventing excessively large policy updates. Specifically, PPO optimizes the clipped surrogate objective

$$\begin{aligned} \mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_n\Big[\min\Big\{ & r_n(\theta)\hat{A}_n, \\ & \text{clip}(r_n(\theta),\, 1-\epsilon,\, 1+\epsilon)\, \hat{A}_n \Big\}\Big], \end{aligned} \tag{22}$$

where $r_n(\theta) = \frac{\pi_\theta(\mathbf{a}_d[n]|\mathbf{s}[n])}{\pi_{\theta_{\text{old}}}(\mathbf{a}_d[n]|\mathbf{s}[n])}$ is the probability ratio between the new and old policies, ϵ is the clipping parameter, and $\hat{A}_n$ denotes the advantage estimate. To reduce variance and improve sample efficiency, the advantage is computed using Generalized Advantage Estimation (GAE),

$$\hat{A}_n = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{n+l}, \quad \delta_n = r[n] + \gamma V(\mathbf{s}[n+1]) - V(\mathbf{s}[n]), \tag{23}$$

where $V(\cdot)$ is the value function, γ is the discount factor, and λ controls the bias–variance tradeoff.

DDPG for Trajectory/Power Control: Given the semantic-augmented state $\mathbf{s}[n]$, we employ DDPG to learn the continuous control policy in an off-policy manner. Specifically, the transition tuple $(\mathbf{s}[n], \mathbf{a}_c[n], r[n], \mathbf{s}[n+1])$ is stored in a replay buffer, and mini-batches are uniformly sampled to update the

critic network $Q_\omega(\mathbf{s}, \mathbf{a}_c)$. For each sampled transition, the bootstrap target is constructed by the target networks as

$$y[n] = r[n] + \gamma\, Q_{\bar{\omega}}\Big(\mathbf{s}[n+1],\, \mu_{\bar{\varphi}}(\mathbf{s}[n+1])\Big), \tag{24}$$

and the critic is trained by minimizing the temporal-difference loss

$$L_c(\omega) = \mathbb{E}\Big[\big(Q_\omega(\mathbf{s}[n], \mathbf{a}_c[n]) - y[n]\big)^2\Big]. \tag{25}$$

The actor is then updated to maximize the critic-evaluated Q-value, while the corresponding target networks are updated via Polyak averaging. At each time slot, PPO outputs the discrete spectrum allocation $\mathbf{a}_d[n]$ (on-policy), and DDPG outputs the continuous control action $\mathbf{a}_c[n]$ (off-policy) based on $\mathbf{s}[n]$.

IV. SIMULATION RESULTS

In this section, simulation results are presented to evaluate the performance of the proposed joint spectrum allocation, trajectory optimization, and power control scheme. The simulation settings are based on the works in [1]. Specifically, SUs are PUs are randomly deployed in two seperate $500\,\mathrm{m} \times 500\,\mathrm{m}$ cells with a C-UAV flying at a fixed altitude of $100\,\mathrm{m}$. The number of SUs is 4. The transmit power of C-UAV ranges from 200mW to 300mW. The sensing window length for mode recognition is $L = 5$. The transition matrix is set as

$$\mathbf{\Pi} = \begin{bmatrix} 0.85 & 0.10 & 0.05 \\ 0.10 & 0.80 & 0.10 \\ 0.10 & 0.10 & 0.80 \end{bmatrix}. \tag{26}$$

Two baseline schemes are introduced for performance comparison. **MA2C-DDPG:** MA2C and DDPG are used for spectrum allocation and trajectory optimization, with the transmit power of the C-UAV fixed at 250 mW. **Hybrid PPO-DDPG:** Hybrid PPO-DDPG framework without the jamming mode recognition module.

V. CONCLUSION

This paper investigated mode-aware anti-jamming communication in dynamic spectrum sharing UAV networks under pattern-switching interference. A semantic-aware framework was developed in which the UAV first recognizes the current jamming pattern using a CNN–GRU based classifier and then incorporates the inferred mode information into the decision-making process. By formulating a semantic-augmented MDP, a hybrid PPO–DDPG reinforcement learning algorithm was proposed to jointly optimize spectrum allocation, UAV trajectory, and transmit power. Simulation results demonstrated that the proposed approach can effectively adapt to dynamic interference environments, achieve faster recovery after jamming mode transitions, and significantly improve both average throughput and communication reliability compared with existing schemes. The proposed framework provides a promising solution for robust UAV communications in adversarial spectrum-sharing environments. Future work will investigate more complex adversarial scenarios, including multiple cooperative jammers and online adaptation of the jamming recognition module.

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